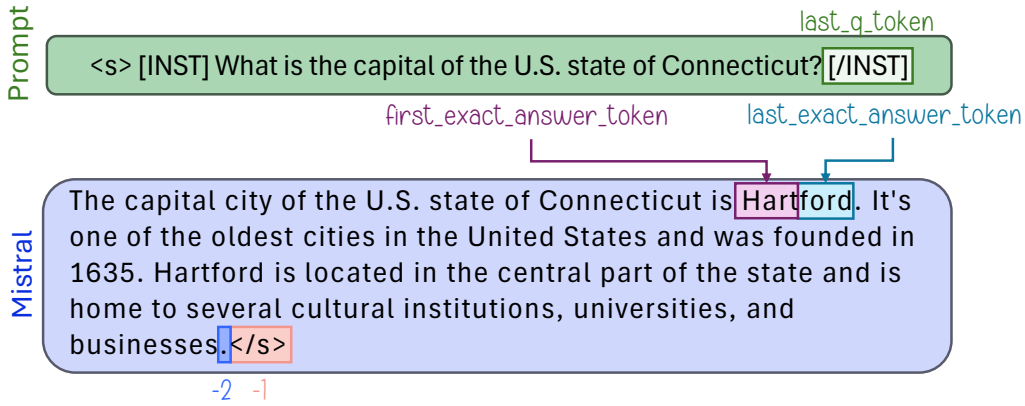
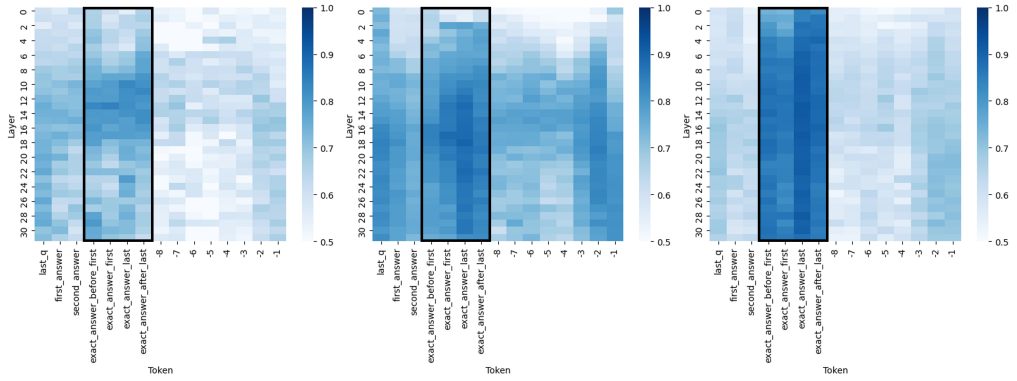


Figure 1: Example for the input and LLM output from the TriviaQA dataset, and the names of the tokens that can be probed.



(a) TriviaQA (b) Winobias (c) Math

Figure 2: AUC values of a probe error detector across layers and tokens, Mistral-7b-instruct. Generation proceeds from left to right, with detection performance peaking at the exact answer tokens.

each error detection method with an "exact answer" version, demonstrating that it often improves performance, especially in probing. Implementation details for detecting the exact answer token are given in Appendix A.2.

3.3 Results

Patterns of truthfulness encoding. We first focus on probing classifiers to gain insights into the internal representations of LLMs. Specifically, we analyze the effects of layer and token selection on the error detection performance of these probing classifiers. By systematically probing all model layers, starting from the last question token to the final generated token, we observe consistent truthfulness encoding patterns. Figure 2 shows AUC metrics of probes across Mistral-7b-Instruct layers and tokens. Middle to later layers often yield the most effective probing results (see Appendix B for more datasets and models), aligning with previous studies on truthfulness encoding (Burns et al., 2022; CH-Wang et al., 2023) and transformer representations (nostalgebraist, 2020; Meng et al., 2022; Geva et al., 2023). Regarding tokens, a strong truthfulness signal appears immediately after the prompt, suggesting that this representation encodes information on the model's general ability to answer the question correctly. This signal weakens as text generation progresses but peaks again at the exact answer tokens. Towards the end of the generation process, signal strength rises again, though it remains weaker than at the exact answer tokens. These patterns are consistent across nearly all datasets and models (see Appendix B), suggesting a general mechanism by which LLMs encode and process truthfulness during text generation.

Table 1: Comparison of error detection techniques using AUC metric, across different models and datasets. The best-performing method is **bolded**. Using exact answer tokens is useful for many cases, especially probing.

	Mistral-7b-Instruct			Llama 3-8b-Instruct		
	TriviaQA	Winobias	Math	TriviaQA	Winobias	Math
Logits-mean	0.60 ±0.009	0.56 ±0.017	0.55 ±0.029	0.66 ±0.005	0.60 ±0.026	0.75 ±0.018
Logits-mean-exact	0.68 ±0.007	0.54 ±0.012	0.51 ±0.005	0.71 ±0.006	0.55 ±0.019	0.80 ±0.021
Logits-min	0.63 ±0.008	0.59 ±0.012	0.51 ±0.017	0.74 ±0.007	0.61 ±0.024	0.75 ±0.016
Logits-min-exact	0.75 ±0.006	0.53 ±0.013	0.71 ±0.009	0.79 ±0.006	0.61 ±0.019	0.89 ±0.018
p(True)	0.66 ±0.006	0.45 ±0.021	0.48 ±0.022	0.73 ±0.008	0.59 ±0.020	0.62 ±0.017
p(True)-exact	0.74 ±0.003	0.40 ±0.021	0.60 ±0.025	0.73 ±0.005	0.63 ±0.014	0.59 ±0.018
Probe @ token						
Last generated [-1]	0.71 ±0.006	0.82 ±0.004	0.74 ±0.008	0.81 ±0.005	0.86 ±0.007	0.82 ±0.016
Before last generated [-2]	0.73 ±0.004	0.85 ±0.004	0.74 ±0.007	0.75 ±0.005	0.88 ±0.005	0.79 ±0.020
End of question	0.76 ±0.008	0.82 ±0.011	0.72 ±0.007	0.77 ±0.007	0.80 ±0.018	0.72 ±0.023
Exact	**0.85** ±0.004	**0.92** ±0.005	**0.92** ±0.008	**0.83** ±0.002	**0.93** ±0.004	**0.95** ±0.027

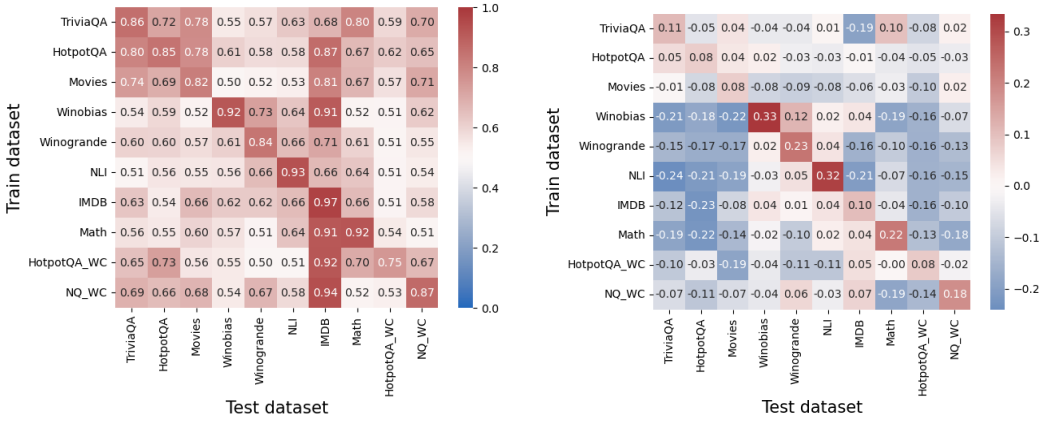
Error Detection Results. Next, we evaluate various error detection methods by comparing their performance with and without the use of exact answer tokens. Table 1 compares the AUC across three representative datasets (additional datasets and models in Appendix B, showing consistent patterns). Here we present results for the last exact answer token, which outperformed both the first exact answer token and the one preceding it, while the token following the last performed similarly. Incorporating the exact answer token improves the different error detection methods in almost all datasets. Notably, our probing technique (bottom line) consistently outperforms all other baselines across the board. While we did not compare all existing error detection methods, the primary conclusion is that information about truthfulness is highly localized in specific generated tokens, and that focusing on exact answer tokens leads to significant improvements in error detection.

4 Generalization Between Tasks

The effectiveness of a probing classifier in detecting errors suggests that LLMs encode information about the truthfulness of their outputs. This supports using probing classifiers for error detection in production, but their generalizability across tasks remains unclear. While some studies argue for a universal mechanism of truthfulness encoding in LLMs (Marks & Tegmark, 2023; Slobodkin et al., 2023), results on probe generalization across datasets are mixed (Kadavath et al., 2022; Marks & Tegmark, 2023; CH-Wang et al., 2023; Slobodkin et al., 2023; Levinstein & Herrmann, 2024)–observing a decline in performance, yet it remains significantly above random chance. Understanding this is essential for real-world applications, where the error detector may encounter examples that significantly differ from those it was trained on. Therefore, we explore whether a probe trained on one dataset can detect errors in others.

Our generalization experiments are conducted between all of the ten datasets discussed in Section 3, covering a broader range of reaslistic task settings than previous work. This breadth of experiments has not been previously explored, and is crucial considering the mixed findings in previous work. We select the optimal token and layer combination for each dataset, train all probes using this combination on other datasets, and then test them on the original dataset. We evaluate generalization performance using the absolute AUC score, defined as $\max(\text{auc}, 1 - \text{auc})$, to also account for cases where the learned signal in one dataset is reversed in another.

Results. Figure 3a shows the generalization results for Mistral-7b-instruct, with similar patterns observed for other LLMs in Appendix C. In this context, values above 0.5 indicate successful generalization. At first glance, the results appear consistent with previous research: most heatmap values exceed 0.5, implying some degree of generalization across tasks. This observation supports the existence of a universal mechanism for decoding truthfulness, since the same linear directions—captured by the probe—encode truthfulness information across many datasets. However, upon closer inspection, it turns out that most of this performance can be achieved by logit-based truthfulness detection,



(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 3: Generalization between datasets, Mistral-7b-instruct. After subtracting the logit-based method's performance, we observe that most datasets show limited or no meaningful generalization.

which only observes the output logits. Figure 3b presents the same heatmap after subtracting results from our strongest logit-based baseline (Logit-min-exact). This adjusted heatmap reveals the probe's generalization rarely exceeds what can be achieved by examining logits alone. This suggests that the observed generalization is not due to a universal internal encoding of truthfulness. Instead, it likely arises from information already available through external features, such as logits. Past evidence for generalization may therefore have been overstated.

Nonetheless, we do observe some successful generalization in tasks requiring similar skills, such as parametric factual retrieval (TriviaQA, HotpotQA, Movies) and common-sense reasoning (Winobias, Wingrande, NLI). This suggests that, although the overall pattern of truthfulness signals across tokens appeared consistent across tasks (as observed in Section 3.3), LLMs have many "skill-specific" truthfulness mechanisms rather than universal ones. However, some patterns remain unexplained, such as the asymmetric generalization from TriviaQA to Math tasks. Overall, our findings indicate that models have a multifaceted representation of truthfulness. The internal mechanisms responsible for solving distinct problem are implemented as different mechanisms (e.g., circuits) within models (Elhage et al., 2021; Olah et al., 2023). Similarly, LLMs do not encode truthfulness through a single unified mechanism but rather through multiple mechanisms, each corresponding to different notions of truth. Further investigation is required to disentangle these mechanisms.

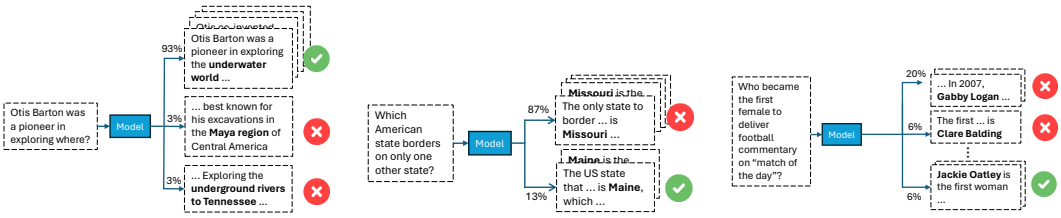
5 Investigating Error Types

Having established the limitations of error detection, we now shift to error analysis. Previously, we explored types of LLM limitations across *different* tasks, noting both commonalities and distinctions in their error representations. In this section, we focus on the types of errors LLMs make in a *specific* task—TriviaQA—which represents factual errors, a commonly studied issue in LLMs (Kadavath et al., 2022; Snyder et al., 2023; Li et al., 2024; Chen et al., 2024; Simhi et al., 2024).

5.1 Taxonomy of Errors

Intuitively, not all mistakes are identical. In one case, an LLM may consistently generate an incorrect answer, considering it correct, while in another case, it could issue a best guess. To analyze errors from the LLM's perspective, we sample $K = 30$ responses at a temperature setting of $T = 1$[4] for each example in the dataset and then analyze the resulting distribution of answers.

[4]We chose $K = 30$ as the overall correctness seemed to plateau around this point; see Appendix D. We found that lower temperatures generally produced less truthful answers across repeated trials.



(a) The LLM mostly answers correctly, but sometimes hallucinates.

(b) The LLM mostly answers incorrectly, but seems to have some knowledge on the correct answer.

(c) The LLM generates many different answers, one of them is the correct one which is generated a small fraction of the resamples.

Figure 4: Different error types in free-form generation, exposed when resampled many times.

Figure 4 illustrates three representative error types. In one (Figure 4a), the model usually gives the correct answer but occasionally make an error, implying correct information is present but sampling may lead to mistakes. In another (Figure 4b, the model often responds incorrectly, though it is capable of providing the right answer, indicating some retained knowledge despite consistently making the same error. In a third type (Figure 4c), the model generates a wide array of mostly incorrect answers, reflecting low confidence in any generated answer.

More generally, we categorize the errors by logging three specific features for each example: (a) the number of different answers generated; (b) the frequency of the correct answer; and (c) the frequency of the most common incorrect answer. These features reveal the following error patterns:

- **(A) Refuses to answer:** The model responds that it cannot answer the question in at least half the cases.
- **(B) Consistently correct:** Answers correctly in at least half of the cases. This category is divided into: (B1) always correct; and (B2) mostly correct with occasional errors.
- **(C) Consistently incorrect:** Consistently generates the same incorrect response in at least half of the cases. Similarly to type B, we subdivide this type into (C1) correct answer is never produced; and (C2) correct answer appears at least once.
- **(D) Two competing:** Generates both correct and incorrect responses at similar rates– difference in rates is 5 or less, and each response is generated at least 5 times.
- **(E) Many answers:** Generates over 10 distinct answers. Like types C and D, Subtypes include (E1) correct answer is never generated; and (E2) correct answer is generated at least once.

This taxonomy covers 96% of the errors in TriviaQA for Mistral-7b-instruct. For more qualitative examples of each type of error, see Appendix D.3. Although some overlap exists between types, our goal is to identify general patterns and explore their connection to the models's internal representations. For a discussion on the design choices of this taxonomy, refer to Appendix D.1. This taxonomy classifies LLM errors based on an extrinsic, behavior-based analysis. Similarly, previous work analyzed repeated samples to assess an LLM's knowledge of the correct answer (Simhi et al., 2024; Gekhman et al., 2024). Our approach is distinct because it also examines the nature of errors that the LLM makes. Furthermore, as we discuss next, we analyze the connection between these behavioral patterns and the model's internal encoding.

5.2 Predicting Error Types

Our taxonomy offers an external, behavioral analysis of LLMs, which we complement by an intrinsic evaluation. We explore whether LLMs encode information on potential error types within their intermediate activations, offering a deeper insight into the underlying mechanisms. To investigate this, we train a probe in a one-to-many setting, where a single probe identifies a specific error type from all others. We use representations extracted from the answers produced via greedy decoding.

Table 2 presents the results. Our findings show that error types can be predicted from the intermediate representations of the greedy decoding generations, suggesting that they may capture not only